

Coding & Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	Smart Lender– Loan Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation of the Smart Lender – Loan Approval Prediction System. The solution uses Machine Learning to predict loan approval based on applicant information. The application is developed using Python, Flask, HTML, and CSS, following clean coding practices, modular design, and project requirements to ensure reliability, maintainability, and ease of future enhancements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/23ata05317/Smart-Lender-Loan-Approval-Prediction-System
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	Loan approval prediction, data preprocessing, Exploratory Data Analysis (EDA), Machine Learning model training, trained ML model integration, responsive web interface, feature scaling, input validation, prediction result display.
Pending / Incomplete Features	User authentication, prediction history, database integration, email notifications
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , then run <code>python app.py</code> and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender – Loan Approval Prediction System successfully integrates a Machine Learning model with a Flask web application to provide accurate loan approval predictions. The project follows a modular architecture with separate folders for templates, static files, dataset, model, and source code, making it easy to maintain and extend with additional features in future versions. The application has been successfully deployed on Render and the complete source code is available on GitHub.